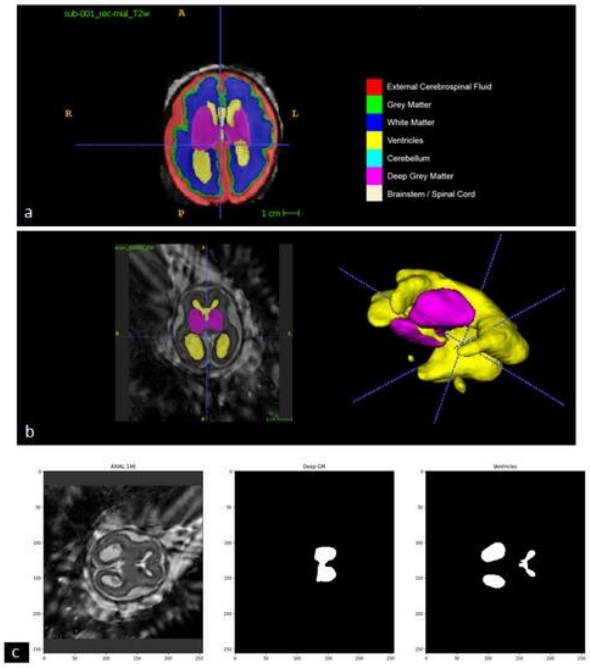
Project 1: Explainable Deep Learning for Fetal MRI Classification



Objective

Develop a deep-learning model to classify fetal brain MRI scans as neurotypical (normal) or affected by spina bifida, using the FeTA dataset. Investigate whether an attention mechanism can highlight the brain regions that contribute most to the model's predictions.

The project combines medical image classification with explainable AI: students should evaluate not only whether the model classifies scans correctly, but also where it looks when making a prediction.

Proposed tasks

1. Dataset preparation: obtain the refined FeTA annotations, identify neurotypical and spina bifida cases, and exclude other pathologies. Record gestational age and whether spina bifida cases were scanned before or after surgery.
2. Preprocessing: normalize 3D T2-weighted MRI volumes, standardize spatial dimensions, and optionally use the provided tissue segmentation maps.
3. Baseline classification: train a 3D CNN to distinguish neurotypical from spina bifida scans. Compare it with a simpler model using extracted MRI features.
4. Attention and explainability: add an attention module or use 3D Grad-CAM to generate heatmaps showing which brain regions contribute to predictions. Overlay the heatmaps on the MRI scans.
5. Evaluation: report sensitivity, specificity, ROC-AUC, F1-score and confusion matrices. Compare attention maps across correctly and incorrectly classified cases.
6. Discussion: assess whether the model identifies plausible anatomical patterns or exploits confounders such as gestational age, acquisition site or surgery status.

Expected outcome

A reproducible classification pipeline, quantitative performance comparison, and MRI visualizations showing the regions associated with the model's decisions.

Important feasibility check: the standard FeTA release provides neurotypical/pathological labels, but not necessarily the diagnosis for every case. The refined dataset paper reports the spina bifida diagnoses, so students must first verify access to the corresponding subject-level diagnosis labels and MRI files. The reported 36 spina bifida cases include pre- and post-surgical cases; these should be tracked separately. With such a small cohort, use subject-level splits and preferably cross-validation, and treat the results as exploratory rather than clinically validated.

Dataset and reference papers

1. FeTA fetal MRI dataset

Public fetal brain MRI reconstructions with tissue segmentation maps and neurotypical/pathological labels. The refined dataset reports 34 neurotypical and 54 pathological cases, including 36 spina bifida cases (pre- and post-surgery). Use only those 36, unless you have cool ideas
Official FeTA dataset and download instructions

Refined dataset paper and diagnosis information

2. Hosna Asma ull et al. , developmental brain MRI review

Developmental Brain Age Estimation From MRI Data: A Systematic Review of Deep Learning Approaches and Open Datasets.

Use this review to explore fetal and developmental MRI datasets, existing deep-learning approaches, and the limitations of available data.

Read the review

3. Subbarao et al. (2025) , ventriculomegaly classification

This paper uses FeTA scans, ventricular segmentation, automated measurements and deep-learning classification. It provides a methodological reference for building a fetal MRI classification pipeline. Read the open-access paper

Project 2 CEBRA for pre-seizure data visualization and prediction.

Objectives

The goal is to investigate whether dimensionality-reduction and representation-learning methods can reveal meaningful patterns in neonatal EEG and support seizure-state classification, compare:

- PCA: linear dimensionality reduction.
- UMAP and t-SNE: nonlinear embedding and visualization.
- CEBRA-Time: contrastive learning of temporal representations.

The project has two components:

1. Unsupervised analysis: Learn and visualize EEG embeddings without using seizure labels during training. Investigate whether normal, pre-ictal, and ictal segments exhibit distinct latent structures.
2. Supervised classification: Train a common downstream classifier on each representation to distinguish normal/interictal, pre-ictal, and ictal EEG.

Methodology

1. **Preprocess EEG** and extract fixed-length windows.
2. Learn embeddings using PCA, UMAP, t-SNE, and CEBRA-Time.
3. Visualize latent spaces and investigate their temporal structure.
4. Train comparable classifiers on each representation.
5. Evaluate using subject-independent splits, ensuring that recordings from the same infant never appear in both training and test sets.
6. Compare performance, embedding stability, and generalization.

Define pre-ictal windows using a prespecified interval before seizure onset. Pre-ictal classification does not by itself demonstrate seizure prediction.

Evaluation

Report balanced accuracy, macro-F1, AUROC, AUPRC, and per-class sensitivity. Where feasible, evaluate seizure-level detection, false alarms per hour, and warning time.

Research question

Does temporal contrastive learning with CEBRA provide more useful and generalizable EEG representations than PCA, UMAP, and t-SNE for neonatal seizure-state classification?

Expected outcome: A reproducible pipeline, embedding visualizations, a classification benchmark, and a critical comparison of the methods.

Resources

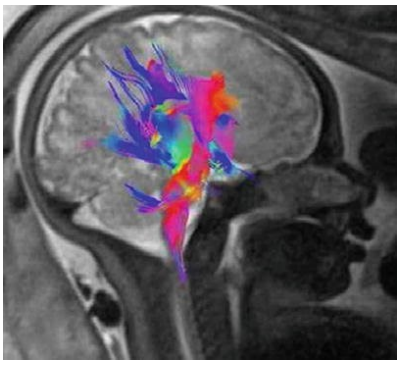
Datasets

- Primary , Helsinki neonatal seizure EEG: https://zenodo.org/records/4940267 Multichannel neonatal EEG with expert seizure annotations.
- External validation , Siena Scalp EEG: https://physionet.org/content/siena-scalp-eeg/1.0.0/ An independent epilepsy dataset from an older population, suitable for exploratory cross-population validation but not direct neonatal validation.

  Look for other epilepsy EEG dataset
- CEBRA repository: https://github.com/AdaptiveMotorControlLab/CEBRA
- EEG processing: https://mne.tools/

Project 3: Explainable Graph Attention Networks

for Preterm Birth Classification Using Neonatal Diffusion MRI



Background. Premature birth can affect the development of cerebral white matter and the organization of structural brain connectivity. Diffusion MRI (dMRI) provides information about white-matter microstructure, while tractography can be used to reconstruct structural connections between brain regions. Although previous studies have investigated preterm-versus-term differences in neonatal structural networks, graph attention networks (GATs) and explainable AI offer opportunities to investigate which brain regions and connections are most informative for classification.

Objective. The aim of this project is to develop an explainable deep-learning framework to distinguish preterm-born infants from term-born controls using neonatal diffusion MRI from the Developing Human Connectome Project (dHCP). The project will investigate whether graph attention networks can identify structural connectivity patterns associated with prematurity and determine which brain regions, white-matter connections, or individual tracts contribute most strongly to the model's predictions.

Methodology

1. **Dataset preparation.** Obtain neonatal diffusion MRI and associated demographic and clinical metadata from dHCP. Define preterm-born infants and term-born controls using gestational age at birth. Record postmenstrual age at scanning and other relevant covariates, and apply appropriate imaging quality-control procedures.
2. **Diffusion processing and feature extraction.** Process the diffusion-weighted images to estimate diffusion tensor metrics, including fractional anisotropy (FA), mean diffusivity (MD), axial diffusivity (AD), and radial diffusivity (RD). Reconstruct white-matter pathways using tractography and derive connectivity matrices and tract-specific measurements for major white-matter bundles where reliably identifiable.
3. **Graph construction.** Represent each infant's structural connectome as a graph, with brain regions as nodes and tractography-derived connections as edges. Node features may include regional diffusion measurements, while edge features may include connection strength, streamline counts, and tract-level diffusion statistics. Compare alternative graph representations, including region-level structural connectomes and tract-specific feature graphs.
4. **Classification models.** Establish conventional machine-learning baselines using diffusion and tract-level features, followed by a graph convolutional network and a graph attention network (GAT). Evaluate whether attention-based message passing improves classification performance and supports more interpretable predictions.

5. **Explainable AI.** Investigate model explanations using attention weights, graph explainers, and feature-attribution methods. Identify the brain regions, connections, and white-matter tracts most relevant to preterm-versus-term classification. Visualize the resulting explanations on anatomical brain representations and compare the consistency of important features across cross-validation folds.
6. **Evaluation and validation.** Evaluate the models using subject-level train/test splits or nested cross-validation, reporting ROC-AUC, balanced accuracy, sensitivity, specificity, and F1-score. Control for potential confounding factors, particularly postmenstrual age at scanning, gestational age at birth, sex, and acquisition-related differences where applicable. Compare the performance of graph-based models with conventional machine learning and diffusion-image-based baselines.

Expected outcomes

The project is expected to deliver a reproducible neonatal diffusion MRI processing and classification pipeline, a comparison of conventional and graph-attention models, and anatomical visualizations of the regions and structural connections associated with model predictions. The results may help characterize which aspects of neonatal white-matter microstructure and structural connectivity are most informative for distinguishing preterm-born infants from term-born controls.

Dataset and references

- **Dataset:** Developing Human Connectome Project (dHCP)
- **Neonatal diffusion MRI processing pipeline:** Bastiani et al., 2019 , Automated processing pipeline for neonatal diffusion MRI in the developing Human Connectome Project
- **Related structural-network classification study:** Preterm-birth alters the development of nodal clustering and neural connection pattern in brain structural network at term-equivalent age
- **Related structural-connectome study:** Network controllability of structural connectomes in the neonatal brain

Project 4: Cross-Modal Brain Registration from MRI to Polarized Light Imaging (PLI)

Project description

Objective. Develop and evaluate a registration pipeline to align MRI brain images with higher-resolution Polarized Light Imaging (PLI) data. The aim is to transfer anatomical information from MRI to PLI while accounting for differences in spatial resolution, contrast, and tissue appearance.

The project will use data from the Oxford Digital Brain Bank, which provides access to human brain imaging resources. Students should first investigate which subjects have both MRI and PLI data available and determine the appropriate registration targets.

Tasks

1. **Dataset exploration:** identify subjects with corresponding MRI and PLI data. Inspect image dimensions, voxel spacing, anatomical coverage, and available metadata. Determine whether the images are already approximately aligned or require substantial registration.
2. **Preprocessing and subsampling:** preprocess the MRI and PLI data and bring them into a common coordinate system where possible. Because MRI generally has much lower spatial resolution than PLI, investigate subsampling or aggregation of PLI data to an MRI-compatible resolution for registration and validation. Preserve the original high-resolution PLI data for the final transformation and visualization.
3. **Registration pipeline:** implement a suitable MRI-to-PLI registration method, starting with rigid and affine transformations and then exploring deformable registration if justified. Consider tools such as ANTs or SimpleITK. Since MRI and PLI have different image contrasts, evaluate cross-modal similarity measures rather than relying exclusively on raw intensity matching.
4. **Registration validation:** investigate how to assess alignment independently of the similarity measure optimized during registration. Possible approaches include:
    - Comparing anatomical landmarks or manually identified corresponding points before and after registration.
    - Measuring landmark target registration error, where suitable landmarks are available.
    - Comparing brain boundaries or tissue interfaces using distance-based metrics.
    - Checking alignment of identifiable anatomical structures across registered slices.
    - Performing sensitivity analyses to determine whether results change substantially with subsampling resolution, initialization, or registration settings.
5. **Visualization and analysis:** display MRI and PLI images side by side and as overlays, checkerboards, or contour plots. Document successful registrations, failure cases, and anatomical regions where alignment remains uncertain.

Expected outcomes

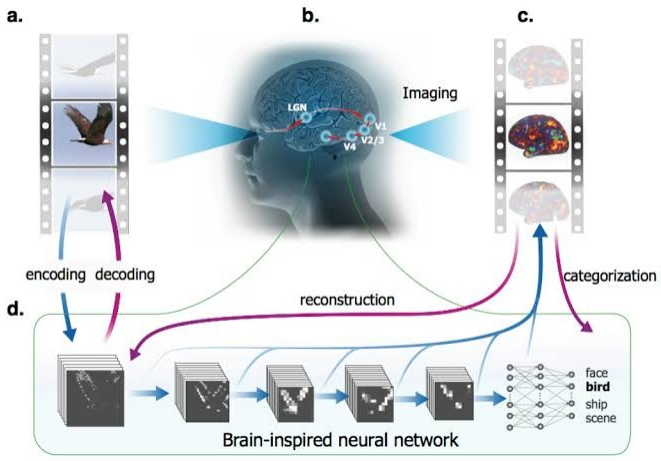
A reproducible cross-modal registration pipeline, quantitative and visual validation of registration quality, and an analysis of how subsampling affects alignment. The project should establish which

anatomical structures can be aligned reliably and where resolution differences or modality-specific contrast limit registration accuracy.

Dataset and resources

- **Dataset:** Oxford Digital Brain Bank (take at least 1 sample of the human corpus callosus)
- **Registration software:** ANTs or SimpleITK or anything that works

Project 5: EEG-to-Image Generation with Generative AI



Objective

Reconstruct an approximate image from EEG recordings of a person viewing natural images, inspired by MindEye's brain-to-image approach but adapted to EEG rather than fMRI.

Proposed methodology

1. Select data: use one THINGS-EEG2 participant and a subset of approximately 500–2,000 training images, with held-out test images.
2. Preprocess EEG: filter, epoch, normalize, and extract spatiotemporal features from the EEG signal.
3. Learn a brain representation: train a lightweight CNN or temporal encoder to predict image embeddings from EEG.
4. Generate images: use the predicted embeddings to condition a pretrained diffusion model or image decoder. Compare semantic reconstruction with lower-level visual reconstruction.
5. Evaluate: compare generated and original images using perceptual similarity, image–text embedding similarity, and qualitative inspection.

Suggested baselines

- A simple EEG-to-image-feature regression model.
- A learned EEG encoder with a frozen pretrained image decoder.
- An optional fine-tuned encoder or diffusion-conditioning model for GPU-equipped teams.

Computational requirements

- The baseline should run with cached pretrained image features and a frozen decoder, without requiring GPU access.
- GPU acceleration is encouraged for training and image generation, but the project should not depend on training a large diffusion model from scratch.
- Use a small data subset and pretrained weights; make model size and data size configurable.

Expected deliverables

- A reproducible notebook for EEG preprocessing, model training, and image generation.
- Side-by-side comparisons of original stimuli and reconstructed images.
- Quantitative and qualitative evaluation.
- A short report on reconstruction quality, generalization, and limitations.

Important: MindEye itself reconstructs images from fMRI, not EEG. The project should use an EEG-specific reconstruction pipeline as its starting point, such as EEG-Image-Reconstruction, rather than attempting to run the original MindEye model unchanged.

Recommended dataset and code



THINGS-EEG2

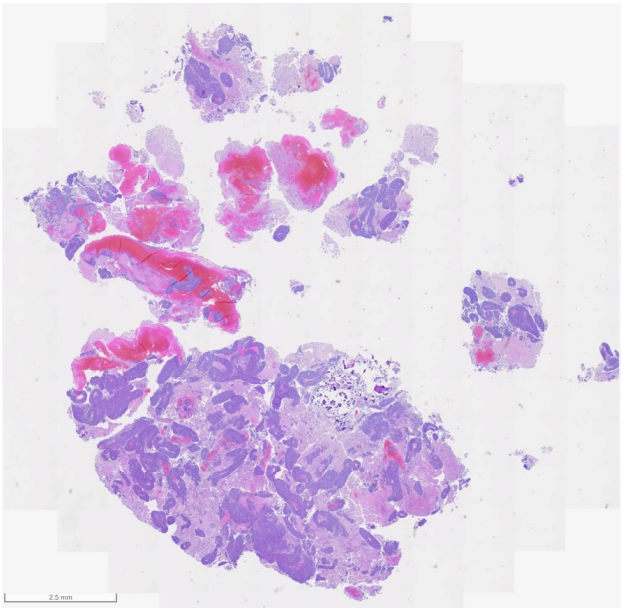
EEG recordings from 10 participants viewing natural images, with stimulus images and preprocessed recordings available. This is a practical starting point for image reconstruction experiments.

- Dataset and download instructions
- EEG-to-image reconstruction code
- Alternative reconstruction implementation: ENIGMA

The full THINGS-EEG2 dataset is large, so the groups should use a single participant and a selected subset of stimuli. The existing reconstruction repositories provide useful reference implementations, but some use extensive pretrained-model checkpoints and multiple GPUs for full experiments.

Project 6: Deep Learning for Histopathological Tissue Segmentation in Brain Metastases

Automated Tissue and Nuclei Segmentation in H&E-Stained Histopathological Images



Objective

Develop an image-processing and deep-learning pipeline to automatically segment tissue regions and cell nuclei in histopathological images of brain metastases originating from lung cancer, using the Brain-Mets-Lung-MRI-Path-Segs dataset.

The project aims to identify tissue fragments, distinguish tissue from background, and quantify cellular characteristics such as nuclei density and spatial distribution. Students should investigate how classical image-processing techniques compare with deep-learning approaches for histopathological segmentation.

The project combines digital pathology, biomedical image processing, and deep learning, with the ultimate goal of extracting quantitative tissue characteristics that could support automated histopathological analysis.

Proposed tasks

1. **Dataset preparation:** obtain and inspect the H&E-stained whole-slide histopathological images provided in the project repository. Understand the SVS format and investigate how tissue fragments are distributed within each slide.
2. **Image preprocessing:** implement colour normalization and colour deconvolution to separate haematoxylin and eosin components. Investigate the effect of staining variability on segmentation performance.
3. **Tissue segmentation:** develop methods to distinguish tissue regions from the glass-slide background. Compare thresholding, colour-based segmentation, and morphological operations.
4. **Nuclei segmentation:** investigate methods for detecting and segmenting individual cell nuclei. Compare watershed, classical image processing, and deep-learning approaches such as U-Net or Cellpose.

5. **Quantitative analysis:** calculate tissue area, nuclei density, nuclei size, and spatial distribution. Generate heatmaps describing the cellularity of different tissue regions.
6. **Evaluation:** manually annotate a subset of image patches and evaluate segmentation performance using Dice coefficient, Intersection over Union (IoU), precision, and recall.
7. **Discussion:** investigate the limitations introduced by staining variability, tissue fragmentation, overlapping nuclei, and the absence of expert pixel-level annotations.

Expected outcome

A reproducible histopathological image-processing pipeline capable of identifying tissue regions and segmenting cell nuclei, together with quantitative maps describing tissue cellularity and spatial heterogeneity.

The final demonstration should display the original histological images alongside the predicted segmentation masks and quantitative cellularity maps.

Dataset and reference papers

Dataset A — Brain-Mets-Lung-MRI-Path-Segs (TCIA)

The project repository provides histopathological H&E whole-slide images (SVS format), together with clinical information in an Excel file. The dataset contains 111 image records corresponding to 103 unique patients, including clinical and pathological characteristics.

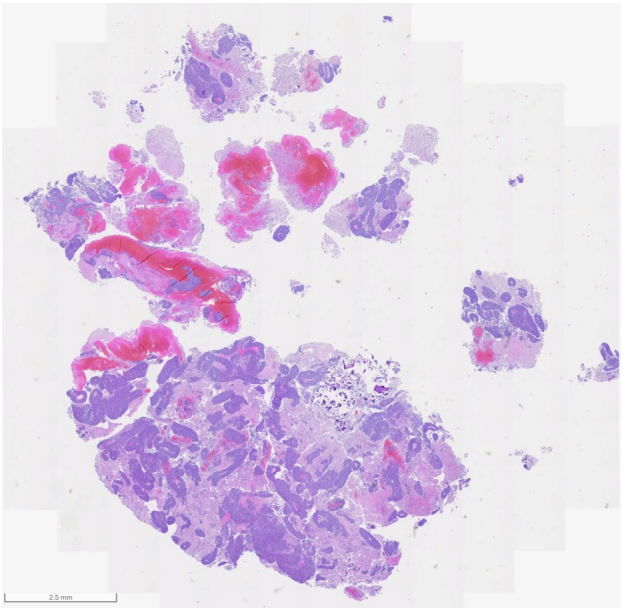
Students can use the clinical information to explore relationships between quantitative histopathological measurements and patient characteristics.

Official TCIA collection and download instructions

Use Histology and Clinical data

Project 7. Learning for Histopathological Classification of Brain Metastases

Predicting Lung Cancer Histological Subtypes from H&E-Stained Tissue Images



Objective

Develop a machine-learning model to classify histological subtypes of lung cancer using H&E-stained histopathological images of brain metastases from the Brain-Mets-Lung-MRI-Path-Segs dataset.

The project aims to investigate whether morphological patterns in histopathological images contain sufficient information to distinguish different lung cancer subtypes, particularly small-cell lung cancer (SCLC), lung adenocarcinoma, and other non-small-cell lung cancer (NSCLC) subtypes.

Students should investigate both conventional machine-learning approaches based on extracted image features and deep-learning models trained directly on image patches.

The project combines digital pathology, computer vision, and supervised learning, with the clinical objective of exploring automated histopathological classification.

Proposed tasks

1. **Dataset preparation:** load the histopathological images and the clinical Excel file provided in the project repository. Associate each image with its corresponding patient ID and histological subtype.
2. **Image preprocessing:** identify tissue regions, remove background areas, normalize staining, and extract smaller image patches from the whole-slide images.
3. **Feature extraction:** investigate handcrafted image descriptors, including colour histograms, texture features, nuclei density, and morphological characteristics.
4. **Baseline classification:** train conventional machine-learning models such as Random Forest, Support Vector Machines (SVM), and XGBoost using the extracted image features.

5. **Deep learning:** implement a CNN-based classification model or use transfer learning with pretrained architectures such as ResNet or EfficientNet. Compare performance with conventional machine-learning approaches.
6. **Model interpretation:** investigate which image regions contribute most to the predictions using Grad-CAM or other explainability techniques. Visualize the morphological patterns associated with different predictions.
7. **Evaluation:** report balanced accuracy, macro-F1 score, sensitivity, specificity, and confusion matrices. Investigate the impact of class imbalance and the limited number of patients.
8. **Discussion:** assess whether histopathological image features can distinguish the selected cancer subtypes and discuss the limitations of automated classification compared with expert pathological assessment.

Expected outcome

A reproducible machine-learning pipeline for histopathological classification, including image preprocessing, patch extraction, feature extraction, and supervised classification.

The final demonstration should allow students to upload a histopathological image or select an image patch and obtain a predicted cancer subtype, together with a visualization highlighting the image regions contributing to the prediction.

Dataset and reference papers

Dataset A — Brain-Mets-Lung-MRI-Path-Segs (TCIA)

The project repository contains H&E-stained whole-slide histopathological images and an accompanying clinical Excel file with patient-level information, including histological subtype, age, sex, EGFR status, ALK status, PD-L1 status, and other clinical characteristics.

The clinical data provide the reference labels for supervised classification.

The dataset includes different histological subtypes, such as adenocarcinoma, SCLC, LCNEC, and other NSCLC subtypes. Students should inspect the class distribution and define suitable classification groups before training their models.

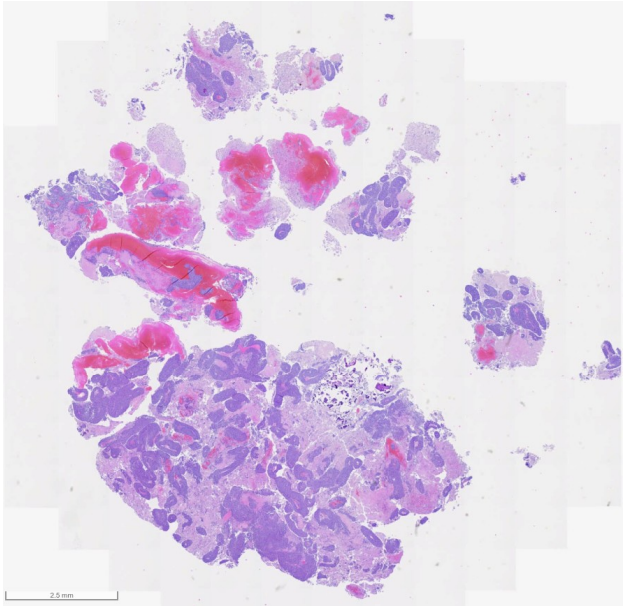
Official TCIA collection (histology, and clinical excel file)

Recommended

- PyTorch
- MONAI

Project 8: Multimodal Machine Learning for Brain Metastases Characterization

Combining MRI Radiomics and Histopathological Image Features to Predict Clinical Characteristics



Objective

Develop a multimodal machine-learning framework combining MRI-derived radiomic features and histopathological image features to investigate the relationship between macroscopic tumour characteristics and microscopic tissue morphology in patients with brain metastases originating from lung cancer.

The project aims to investigate whether combining information from MRI and histopathological images improves the prediction of clinical and pathological characteristics compared with using either imaging modality independently.

Students should extract quantitative features from histopathological images and combine them with the MRI radiomic features and clinical information provided in the project repository.

Potential prediction targets include lung cancer histological subtype, EGFR mutation status, and other available clinical characteristics.

The project combines medical image analysis, radiomics, digital pathology, and multimodal machine learning, with the objective of exploring complementary information provided by different imaging modalities.

Proposed tasks

1. **Dataset preparation:** load the MRI radiomic features, histopathological images, and clinical Excel file provided in the project repository. Establish the correspondence between accession numbers and patient IDs.

2. **MRI feature analysis:** investigate the available radiomic features, including shape, first-order intensity, and texture descriptors (GLCM, GLRLM, GLSZM, and GLDM). Perform feature normalization and investigate correlations between radiomic features.
3. **Histopathological feature extraction:** extract image patches from the H&E whole-slide images. Calculate quantitative features such as tissue cellularity, nuclei density, colour distributions, and texture descriptors. Alternatively, use pretrained CNN architectures to generate deep-learning image embeddings.
4. **Feature selection:** investigate dimensionality reduction and feature-selection methods, such as PCA, mutual information, and LASSO, to identify the most informative features from each modality.
5. **Unimodal classification:** develop baseline machine-learning models using MRI radiomics alone and histopathological features alone. Compare Random Forest, SVM, and gradient-boosting classifiers.
6. **Multimodal integration:** combine MRI radiomic and histopathological features using early fusion (feature concatenation) or late fusion (combining predictions from separate models). Investigate whether multimodal integration improves classification performance.
7. **Clinical association analysis:** investigate relationships between the extracted imaging features and available clinical variables, such as histological subtype, EGFR status, ALK status, PD-L1 status, and number of brain metastases.
8. **Evaluation:** compare MRI-only, histology-only, and multimodal models using balanced accuracy, macro-F1 score, ROC-AUC, and appropriate cross-validation strategies. Ensure that data from the same patient are never present in both training and testing sets.
9. **Discussion:** evaluate the potential advantages of combining MRI and histopathology, the limitations of the relatively small dataset, missing clinical information, and the challenges associated with multimodal learning.

Expected outcome

A reproducible multimodal machine-learning pipeline integrating MRI radiomic features and histopathological image features.

The final demonstration should compare the performance of MRI-only, histology-only, and combined models, and provide visualizations identifying the most informative features from each modality.

The project should investigate whether microscopic tissue characteristics provide complementary information to MRI-derived radiomics when characterizing brain metastases.

Dataset and reference papers

Dataset A — Brain-Mets-Lung-MRI-Path-Segs (TCIA)

The project repository provides:

- H&E-stained whole-slide histopathological images.
- An Excel file containing patient-level clinical and pathological characteristics.

- MRI-derived radiomic features extracted from T1CE and FLAIR images. (either use the original MRI or the metrics in the Excel file)
- Patient IDs and accession numbers to establish correspondence between the different data sources.

The clinical information includes histological subtype, EGFR and ALK mutation status, PD-L1 status, number of brain metastases, lesion size, Karnofsky Performance Status (KPS), and Sperduto GPA.

Software references

- PyRadiomics
- PyTorch
- MONAI
- Scikit-learn

Project 9: EEG-Based Assistive Communication for People with ALS

P300 Speller: From EEG Signals to Text Communication

Objective

Develop a machine-learning model to decode the P300 event-related potential (ERP) from EEG recordings of people with amyotrophic lateral sclerosis (ALS), using the BNCI 2014-008 dataset.

The project aims to simulate an assistive communication system in which a user selects letters by focusing attention on a visual target. Students should investigate whether EEG classification can reliably distinguish target from non-target stimuli and reconstruct the intended characters.

The project combines EEG signal processing, machine learning, and assistive technology: students should evaluate not only classification accuracy, but also the practical communication performance of the resulting BCI.

Proposed tasks

1. Dataset preparation: obtain the EEG recordings and stimulus annotations. Identify target and non-target epochs and reconstruct the experimental spelling trials.
2. Preprocessing: apply appropriate EEG filtering, baseline correction, and epoch extraction. Investigate whether additional artifact rejection improves performance.
3. Baseline classification: train LDA, logistic regression, and SVM models to distinguish target from non-target EEG responses.
4. Deep learning: implement a compact CNN or EEGNet architecture and compare it with conventional classifiers.
5. Spelling reconstruction: combine the predicted target probabilities across stimulus repetitions to select the most likely character in the 6×6 matrix.
6. Evaluation: report balanced accuracy, sensitivity, specificity, ROC-AUC, character accuracy, and simulated information transfer rate. Investigate how performance changes with the number of stimulus repetitions.
7. Discussion: assess the limitations of copy-spelling experiments, the small number of participants, and the difference between offline decoding and real-time communication.

Expected outcome

A reproducible P300 classification pipeline, a virtual spelling interface driven by recorded EEG, and a quantitative evaluation of the trade-off between spelling accuracy and communication speed.

The final demonstration should allow students to select a predefined message from EEG recordings and compare the reconstructed text with the intended message.

Dataset and reference papers

1. Dataset A — BNCI 2014-008 P300 dataset (ALS patients)

EEG recordings from 8 people with ALS performing a P300 visual spelling task using a 6×6 character matrix. The dataset contains 8 EEG channels recorded at 256 Hz, with target and non-target annotations.
NEMAR+1

Official dataset and download instructions

MOABB dataset documentation

2. Riccio et al. (2013) — Attention and P300-based BCI performance in people with ALS

The original study investigates attention and P300-based BCI performance in people with ALS. It provides the clinical and experimental context for the dataset.

Read the paper

Project 10: Dry-Electrode EEG for Assistive Communication

Can We Build a Reliable P300 Speller Without Conventional Wet Electrodes?

Objective

Develop and evaluate an EEG-based P300 classification system using recordings from people with ALS, investigating whether dry electrodes can provide sufficient signal quality for assistive communication.

The project uses the BigP3BCI Study N dataset, which compares dry and wet electrode configurations in people with ALS. Students should investigate whether differences in electrode technology affect P300 detection and spelling performance.

The project combines EEG classification with a practical engineering challenge: reducing the complexity of EEG acquisition while preserving reliable BCI performance.

Proposed tasks

1. Dataset preparation: obtain the Study N EEG recordings and identify participants, recording sessions, electrode configurations, and stimulus annotations.
2. Preprocessing: standardize the EEG preprocessing pipeline across dry and wet recordings. Investigate whether differences in noise and signal quality require different preprocessing strategies.
3. Baseline classification: implement LDA, SVM, and logistic regression models for target versus non-target classification.
4. Electrode comparison: compare P300 classification performance between dry and wet electrode recordings. Analyze whether the differences are consistent across participants.
5. Spelling reconstruction: reconstruct intended characters using the predicted target probabilities and compare the resulting spelling performance.
6. Evaluation: report balanced accuracy, ROC-AUC, character accuracy, and simulated information transfer rate. Compare the performance of dry and wet electrode configurations.
7. Discussion: assess whether dry electrodes could reduce setup complexity in assistive communication systems, considering the small cohort and differences in recording sessions.

Expected outcome

A reproducible comparison of dry- and wet-electrode P300 classification, quantitative performance evaluation, and an analysis of whether dry electrodes retain sufficient EEG information for assistive communication.

Dataset and reference papers

1. Dataset C — BigP3BCI Study N: dry versus wet electrodes

EEG recordings from 8 participants with ALS using a P300-based speller with a 9×8 character grid. The dataset contains 16-channel EEG recorded at 256 Hz, with two sessions comparing dry and wet electrode configurations.
NEMAR+1

Official dataset and download instructions

2. BigP3BCI — Large-scale P300 BCI dataset

The broader BigP3BCI collection provides a context for studying P300-based communication systems and their variability across participants and recording conditions.

Explore the BigP3BCI collection

Project 11: Cross-Subject EEG Classification for Assistive Communication

Can a P300 BCI Work for a New User Without Extensive Calibration?

Objective

Develop a subject-independent EEG classification model for P300-based assistive communication using recordings from people with ALS.

The project aims to investigate whether a machine-learning model trained on some participants can reliably classify EEG responses from previously unseen participants. Students should compare conventional subject-specific calibration with population-based and transfer-learning approaches.

The project combines EEG classification with a central challenge in practical BCI deployment: reducing the need for lengthy calibration sessions for people with severe motor disabilities.

Proposed tasks

1. Dataset preparation: obtain the BNCI 2014-008 and/or BigP3BCI recordings. Organize the data by participant, session, and stimulus condition.
2. Preprocessing: standardize EEG filtering, epoch extraction, and feature scaling. Ensure that all preprocessing parameters are estimated from the training data only.
3. Subject-specific baseline: train and evaluate separate classifiers for each participant using within-subject training and testing.
4. Subject-independent classification: train a pooled model using several participants and evaluate it on participants excluded from training.
5. Calibration analysis: investigate whether a small number of labeled trials from a new participant can improve classification performance.
6. Evaluation: report balanced accuracy, ROC-AUC, and character accuracy. Compare subject-specific, subject-independent, and calibrated models across participants.
7. Discussion: assess the limitations of the small number of participants and the implications of subject variability for assistive BCI deployment.

Expected outcome

A reproducible comparison of subject-specific and subject-independent EEG classifiers, quantitative evidence on the value of calibration, and an analysis of the feasibility of population-based P300 BCI systems.

Dataset and reference papers

1. Dataset A — BNCI 2014-008 P300 dataset (ALS patients)

Eight ALS participants performing a visual P300 spelling task. Suitable for a compact cross-subject classification experiment.
NEMAR+1

Official dataset

2. Dataset B — BigP3BCI

A larger P300 BCI resource suitable for comparing participant variability and calibration approaches across multiple studies.

PhysioNet dataset

Project 12: EEG-Based Movement Intention Detection for Assistive Control

Motor Imagery Classification for Hands-Free Control of Assistive Devices

Objective

Develop a machine-learning model to decode imagined movement from EEG recordings, using the BCI Competition IV Dataset 2a.

The project aims to classify imagined left-hand, right-hand, foot, and tongue movements and investigate how the resulting EEG decoder could be used as an input method for assistive technologies.

The project combines motor-imagery EEG analysis with assistive control: students should evaluate whether the model can reliably identify imagined movement classes and translate them into commands for a simulated device.

Proposed tasks

1. Dataset preparation: obtain the BCI Competition IV Dataset 2a and identify the EEG recordings and event annotations for the four motor-imagery classes.
2. Preprocessing: apply appropriate bandpass filtering and epoch extraction. Investigate the use of common spatial patterns (CSP) for spatial feature extraction.
3. Baseline classification: train LDA, SVM, and logistic regression models using CSP features.
4. Deep learning: implement a compact EEGNet model and compare it with conventional machine-learning approaches.
5. Assistive control: map the predicted motor-imagery classes to commands in a simulated cursor, wheelchair, or smart-home controller. Include a neutral state and a confidence threshold to avoid unreliable commands.
6. Evaluation: report balanced accuracy, macro-F1, confusion matrices, and simulated command error rates. Evaluate performance across participants.
7. Discussion: assess the limitations of motor-imagery BCI systems and discuss whether the results could generalize to people with severe motor impairments.

Expected outcome

A reproducible motor-imagery classification pipeline, quantitative comparison of conventional and deep-learning models, and a simulated assistive controller driven by EEG predictions.

The final demonstration should allow students to control a virtual device using recorded EEG.

Dataset and reference papers

1. BCI Competition IV Dataset 2a

A public motor-imagery EEG dataset containing recordings from 9 participants, 22 EEG channels, and four motor-imagery classes.
BCI World

Official BCI Competition IV dataset

2. EEGNet: A compact convolutional neural network for EEG-based BCIs

A methodological reference for designing compact deep-learning models for EEG classification.

Lawhern et al. (2018), EEGNet

Project 13: Explainable EEG-Based Assistive Communication

Understanding Which EEG Patterns Drive P300-Based Spelling Decisions

Objective

Develop an explainable machine-learning model for P300-based assistive communication using EEG recordings from people with ALS.

The project aims to investigate whether a classifier identifies physiologically plausible P300 responses or relies on irrelevant features, recording artifacts, or participant-specific characteristics.

The project combines EEG classification with explainable AI: students should evaluate not only how accurately the model identifies target stimuli, but also which channels, time windows, and EEG patterns contribute to its predictions.

Proposed tasks

1. Dataset preparation: obtain the BNCI 2014-008 or BigP3BCI EEG recordings and identify target and non-target trials.
2. Preprocessing: apply EEG filtering, epoch extraction, and appropriate baseline correction. Preserve the temporal structure of the EEG signals.
3. Baseline classification: train an LDA or SVM classifier using EEG features and compare it with a compact CNN.
4. Explainability: use feature importance, channel ablation, temporal ablation, or integrated gradients to identify the EEG channels and time windows contributing to classification.
5. Visualization: generate EEG time-course plots, channel importance maps, and scalp topographies showing the discriminative patterns.
6. Evaluation: report balanced accuracy, ROC-AUC, sensitivity, specificity, and participant-level performance. Compare explainability results across correctly and incorrectly classified trials.
7. Discussion: assess whether the identified patterns are consistent with expected P300 responses and investigate the possibility of confounding effects or data leakage.

Expected outcome

A reproducible P300 classification pipeline, quantitative performance comparison, and EEG visualizations explaining which signal patterns contribute to the model's decisions.

Students should demonstrate whether the classifier is using plausible neurophysiological information rather than relying on artifacts or spurious correlations.

Dataset and reference papers

1. Dataset A — BNCI 2014-008 P300 dataset (ALS patients)

EEG recordings from 8 participants with ALS performing a P300 visual spelling task. The dataset contains target and non-target annotations and is suitable for investigating the temporal and spatial patterns underlying P300 classification.

NEMAR+1

Official dataset

2. Dataset B — BigP3BCI

A larger P300 BCI resource that can support comparisons across participants and recording conditions.

PhysioNet dataset

3. Lawhern et al. (2018) — EEGNet

A compact CNN architecture designed for EEG-based brain–computer interfaces. Useful as a deep-learning baseline for P300 classification.

Read the paper

Project: Kamil Kuwala test (only one group) Evaluating Brain Tumor Segmentation Models on a New Brain Metastasis Dataset

Project description

Objective. Evaluate the generalizability of an existing brain tumor segmentation model on an independent dataset of brain metastases originating from lung cancer. The project will use the publicly available Brain-Mets-Lung-MRI-Path-Segs dataset from The Cancer Imaging Archive (TCIA).

The dataset contains 111 MRI studies from 103 patients, including contrast-enhanced T1-weighted and FLAIR images with tumor segmentations. The annotations were manually delineated and verified by a board-certified neuroradiologist, making the dataset suitable for evaluating segmentation performance against expert annotations. The scans were acquired across different scanners and magnetic field strengths, providing an opportunity to investigate model generalization. Dataset details.

Tasks

Core task , required for all groups

1. Obtain and explore the brainmets TCIA dataset (NOT THE ENTIRE TCIA), identifying available MRI sequences, segmentation masks, and cases suitable for evaluation.
2. Run the existing segmentation model from Kamil Kuwala's MRI Analyzer repository on the new dataset.
3. Compare the predicted masks with the expert annotations using metrics such as Dice similarity coefficient, Hausdorff distance, and sensitivity.
4. Visualize representative successes and failures, and investigate how performance varies across patients, MRI sequences, and acquisition characteristics.

Extension , for larger groups

If the core evaluation is completed successfully, investigate a lightweight approach to improve performance on the new dataset without training the entire model from scratch. Possible directions include:

- **Pseudo-transfer learning:** adapt feature representations or selected model components to the new dataset.
- **Inference-time adaptation or intensity normalization:** investigate whether simple adjustments improve performance across scanners.
- **Limited fine-tuning:** if feasible, fine-tune selected layers using a small subset of annotated cases and evaluate on held-out patients.
- **Error analysis:** investigate whether segmentation errors are associated with tumor size, lesion location, or image quality.

Any adaptation must be evaluated on patients not used for training or tuning.

Dataset and code

- **Dataset:** Brain-Mets-Lung-MRI-Path-Segs , TCIA
- **Existing model and code:** MRI Analyzer , GitHub

Group organization

This is a **group project**, so please find other students interested in working together.

- **Two students:** focus on testing the existing model on the new dataset, evaluating segmentation performance, and documenting the results. One student can also act as a mentor for a larger group.
- **Three or more students:** complete the evaluation and add an extension, such as pseudo-transfer learning, limited fine-tuning, or systematic error analysis.

The goal is to produce a reproducible evaluation, a clear comparison with expert annotations, and, if time permits, a simple method for improving model generalization to a new clinical dataset.